

Practice by Standard: Number Sense

Standard Set 1.0: Students understand the place value of whole numbers.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** How is five thousand forty-one written in standard form? **(3NS1.1)**

A 5401
B 5410
C 5014
D 5041

- 2** Which set of numbers is in order from least to greatest? **(3NS1.2)**

F 211, 256, 308, 347
G 211, 308, 347, 256
H 308, 211, 347, 256
J 347, 308, 256, 211

- 3** Which number has a 5 in the hundreds place and a 1 in the ones place? **(3NS1.3)**

A 1565
B 5516
C 5615
D 6551

QUICK Review

STRATEGY Write the digits of the number in a place-value chart.

How would you read each number in the answer choices?

For more help with writing numbers in standard form, see page 25.

STRATEGY Compare the digits in each place value to write the number from least to greatest.

Which is the smallest number? Which is the greatest number?

For more help with ordering whole numbers, see page 38.

READING HINT The *ones* place is the right most digit. The *hundreds* place is the third digit from the right.

Find the number that has a 1 in the first column and a 5 in the third column.

For more help with identifying place value, see page 24.

QUICK Practice

- 4 Gina has 329 stickers in her collection. Which of these equals 329? (3NS1.5)
- F** $300 + 200 + 90$
G $300 + 20 + 9$
H $3 + 20 + 900$
J $3 + 2 + 9$

- 5 The highest point in California is Mt. Whitney. It has an elevation of 4418 meters. What is this elevation rounded to the nearest hundred? (3NS1.4)
- A** 4000 **C** 4410
B 4400 **D** 4420

- 6 Which digit is in the hundreds place in the number 8312? (3NS1.3)
- F** 1 **H** 3
G 2 **J** 8

- 7 How is three thousand one hundred ninety-five written in standard form? (3NS1.1)
- A** 3195 **C** 3109
B 3159 **D** 3105

QUICK Review

STRATEGY Write each digit as the product of the digit and the place value that it holds.

How many hundreds are there in 329?
How many tens? How many ones?

For more help with writing numbers in expanded notation, see page 24.

STRATEGY Look at the number in the tens place. If it is 5 or greater, round up to the next hundred. Otherwise, round down.

What is the digit in the tens place?

For more help with rounding whole numbers, see page 45.

READING HINT The *hundreds* digit is the third digit from the right.

The number has a ones, tens, hundreds, and thousands digit. Which is the hundreds digit?

For more help with identifying place value, see page 24.

STRATEGY Write the digits of the number in the appropriate places.

How would read each number in the answer choices?

For more help with writing numbers in standard form, see page 25.

Practice on Your Own

Standard Set 1.0

- 8** Which digit in 1594 is in the tens place? (3NS1.3)

F 1 **H** 5
G 4 **J** 9

- 9** The four high scores on a video game are shown in the table.

Video Game Scores	
	5679
	5880
	5103
	5334

Which of the following shows the high scores in order from greatest to least? (3NS1.2)

- A** 5103, 5334, 5679, 5880
B 5880, 5334, 5679, 5103
C 5880, 5679, 5334, 5103
D 5880, 5679, 5103, 5334
- 10** Which number has a 7 in the thousands place and a 3 in the tens place? (3NS1.3)
- F** 7335 **H** 7357
G 7353 **J** 7513
- 11** There are 528 students. Which of these equals 528? (3NS1.5)
- A** $500 + 200 + 80$
B $500 + 20 + 8$
C $5 + 20 + 800$
D $5 + 2 + 8$

- 12** Which digit is in the tens place in the number 3452? (3NS1.3)

F 2 **H** 4
G 3 **J** 5

- 13** What is 7139 rounded to the nearest hundred? (3NS1.4)

A 7200 **C** 7100
B 7140 **D** 7000

- 14** The boys basketball team scored a total of 745 points this season. Which of these equals 745? (3NS1.5)

F $700 + 400 + 50$
G $7 + 40 + 500$
H $700 + 40 + 5$
J $7 + 4 + 5$

- 15** How is nine thousand eighty-one written in standard form? (3NS1.1)

A 9081
B 9018
C 9801
D 9810

- 16** Which digit is in the hundreds place in the number 9781? (3NS1.3)

F 9 **H** 7
G 8 **J** 1

Practice by Standard: Number Sense

Standard Set 2.0: Students calculate and solve problems involving addition, subtraction, multiplication, and division.

DIRECTIONS

Choose the best answer.

QUICK Practice

1 $4000 - 1256 =$ (3NS2.1)

- A** 2744
- B** 2734
- C** 2646
- D** 2644

- 2** There are 8 teams in a volleyball tournament. Each team has 12 players. How many players are there altogether? (3NS2.4)

- F** 4 **H** 84
- G** 20 **J** 96

- 3** Abe did this division problem.

$$420 \div 21 = 20$$

Which problem could he do to check his answer? (3NS2.3)

- A** $21 + 20 = \square$
- B** $21 - 20 = \square$
- C** $21 \times 20 = \square$
- D** $21 \div 20 = \square$

QUICK Review

STRATEGY Use borrowing to subtract the numbers.

How can you borrow from larger place values to subtract the numbers?

For more help with subtracting whole numbers, see page 135.

READING HINT Read the problem carefully to see which operation is needed.

What operation can be used to perform repeated addition?

For more help with solving multiplication problems, see page 157.

STRATEGY What operation can be used to “undo” division?

Division and _____ are inverse operations.

For more help with using inverse relationships, see page 258.

QUICK Practice

- 4 At a teacher conference, 228 teachers were divided evenly into groups of 6. How many groups of teachers were there?

(3NS2.5)

F 38

H 116

G 42

J 222

- 5 What number can be multiplied by 8012 to give the answer 8012?

(3NS2.6)

A 0

C 2

B 1

D 10

- 6 Mrs. Chan bought 5 picture frames. All the picture frames were the same price. The total cost was \$80. How much money did each picture frame cost?

(3NS2.7)

F \$400

H \$16

G \$18

J \$12

- 7 Which number is 8 more than 3268? (3NS2.1)

A 3276

C 3260

B 3266

D 3256

QUICK Review

READING HINT Read the problem carefully to see which operation is needed.

Divide the number of teachers by the number in each group to find the total number of groups.

For more help with solving division problems, see page 642.

STRATEGY Use the properties of 0 and 1 to solve the problem.

What is the result when you multiply a number by 1?

For more help with using the properties of 0 and 1, see page 186.

READING HINT Read the problem carefully to see which operation is needed.

This problem can be solved by dividing \$80 by 5.

For more help with determining unit cost, see page 650.

STRATEGY Use addition to find the number.

What is the result when you add 8 to 3268?

For more help with finding the sum of two whole numbers, see page 92.

Practice on Your Own

Standard Set 2.0

- 8 $7 \times 9 =$ (3NS2.2)

F 56 H 61
G 59 J 63

- 9 Look at the number sentence below.

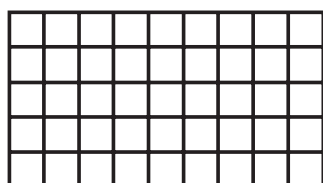
$$28 + \square = 61$$

Which number will make the number sentence true? (3NS2.1)

A 23 C 31
B 27 D 33

- 10 The figure below is a model for the multiplication sentence.

$$5 \times 9 = 45$$



Which division sentence is modeled by the figure? (3NS2.3)

F $36 \div 4 = 9$ H $42 \div 7 = 6$
G $45 \div 5 = 9$ J $45 \div 9 = 5$

- 11 What number can be added to 456 to give the answer 456? (3NS2.6)

A 0 C 10
B 1 D 456

- 12 On Thursday, 325 people visited the park. Four times as many people visited on Friday than on Thursday. How many people visited the park on Friday? (3NS2.4)

F 1200
G 1250
H 1300
J 1325

- 13 The third graders went on a field trip in 4 vans. Each van held 14 students. How many students went on the field trip? (3NS2.4)

A 56 C 48
B 52 D 44

- 14 Roberto had \$20. He paid \$5 for a ticket to the carnival. While there, he bought \$6 worth of ride tickets. What amount of money did Roberto have left? (3NS2.8)

F \$8 H \$10
G \$9 J \$11

- 15 Mikayla did this subtraction problem.

$$1856 - 1524 = 332$$

Which problem could she do to check his answer? (3NS2.3)

A $332 + 1524 =$
B $1524 - 332 =$
C $332 \times 1524 =$
D $1524 \div 332 =$

Practice by Standard: Number Sense

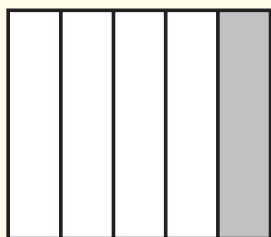
Standard Set 3.0: Students understand the relationship between whole numbers, simple fractions, and decimals.

DIRECTIONS

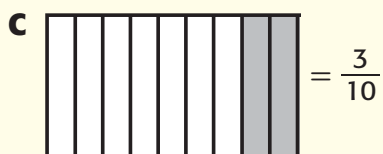
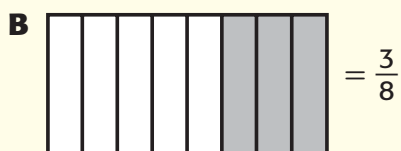
Choose the best answer.

QUICK Practice

- 1 The rectangle shows $\frac{1}{5}$ shaded.



Which fractional part of a rectangle below is equal to $\frac{1}{5}$? (3NS3.1)



QUICK Review

STRATEGY Find a fraction that is equivalent to one-fifth.

Which fraction can you divide the numerator and denominator by the same number to get $\frac{1}{5}$?

$\frac{1}{5}$ ← numerator
 $\frac{1}{5}$ ← denominator

For more help with comparing fractions using models, see page xxx.

QUICK Practice

2 $\frac{1}{6} + \frac{4}{6} =$ (3NS3.2)

F $\frac{5}{6}$

G $\frac{2}{3}$

H $\frac{1}{2}$

J $\frac{5}{12}$

- 3 Jenny has \$20 to buy a CD that costs \$12.38. How much change should she receive? (3NS3.3)

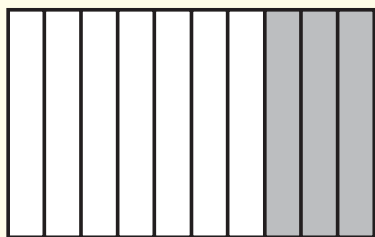
A \$8.62

B \$8.52

C \$7.62

D \$7.52

- 4 Ravi shaded $\frac{3}{10}$ of the figure.



Which decimal equals $\frac{3}{10}$?

(3NS3.4)

F 0.31

G 0.3

H 0.1

J 0.03

QUICK Review

STRATEGY To add fractions with the same denominator, add the numerators and keep the old denominator.

Sketch a rectangle and divide it into 6 equal parts. Shade 1 part. Then shade 4 more parts. How much of the rectangle is shaded?

For more help with adding fractions, see page 528.

READING HINT The amount of change received back is the difference between the starting money and the cost of the item.

Write \$20 as \$20.00. Then use borrowing to find $\$20.00 - \12.38 .

For more help with subtracting money amounts, see page 114.

STRATEGY Find the decimal that is equivalent to $\frac{3}{10}$.

How would you read the fraction $\frac{3}{10}$?
Which decimal reads the same way?

For more help with understanding the equivalence between fractions and decimals, see page 554.

Practice on Your Own

Standard Set 3.0

- 5** A pizza was divided into eighths. Marco ate $\frac{2}{8}$ of the pizza. Peggy ate $\frac{2}{8}$ of the pizza. Derek ate $\frac{3}{8}$ of the pizza. How much of the pizza was left? (3NS3.2)

A $\frac{1}{8}$ **C** $\frac{3}{8}$
B $\frac{2}{8}$ **D** $\frac{4}{8}$

- 6** Dina compared the prices of two book bags. The table below shows the prices.

Cost of Book Bags	
Brand	Cost
A	\$14.73
B	\$12.96

How much more does Brand A cost than Brand B? (3NS3.3)

F \$1.67
G \$1.73
H \$1.77
J \$1.83

- 7** $\frac{2}{8} + \frac{5}{8} =$ (3NS3.2)

A $\frac{3}{16}$ **C** $\frac{7}{16}$
B $\frac{3}{8}$ **D** $\frac{7}{8}$

- 8** Joey bought the three items listed in the table below.

Item	Cost
Swim Goggles	\$8.75
Raft	\$5.60
Beach Ball	\$3.29

What was the total cost of these three items? (3NS3.3)

F \$17.64
G \$16.82
H \$14.35
J \$14.11

- 9** $\frac{2}{8} + \frac{3}{8} =$ (3NS3.2)

A $\frac{1}{8}$ **C** $\frac{4}{8}$
B $\frac{3}{8}$ **D** $\frac{5}{8}$

- 10** Raven has \$10 to buy lunch. Her lunch total comes to \$6.53. How much change should she receive? (3NS3.3)

F \$3.37
G \$3.47
H \$4.37
J \$4.47

- 11** Which decimal equals $\frac{7}{10}$? (3NS3.4)

A 0.07
B 0.17
C 0.7
D \$0.710

Practice by Standard: Algebra and Functions

Standard Set 1.0: Students select appropriate symbols, operations, and properties to represent, describe, simplify, and solve simple number relationships.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** Nia bought 48 apples packed equally into 6 bags. Which number sentence shows how to find the number of apples in each bag?

(3AF1.1)

- A** $48 \times 6 = \square$ **C** $48 - 6 = \square$
B $48 \div 6 = \square$ **D** $48 + 6 = \square$

- 2** Which number makes this number sentence true? (3AF1.2)

$$7 + 5 = \square \times 3$$

- F** 1 **H** 3
G 2 **J** 4

- 3** Which sign goes in the box to make the number sentence true?

(3AF1.3)

$$12 \square 5 = 60$$

- A** + **C** ×
B - **D** ÷

QUICK Review

READING HINT Read the problem statement carefully to see which operation is needed to solve it.

You know the total number of apples and the number of bags. Which operation will tell you how many apples are in each bag?

For more help with modeling situations with number sentences, see page 321.

STRATEGY Eliminate choices first that do not make the number sentence true.

Substitute each answer choice into the box. Then, check to see if a true number sentence is formed.

For more help with solving number sentences, see page 139.

STRATEGY Eliminate choices first that do not make the number sentence true.

Substitute each answer choice into the box. Then, check to see if a true number sentence is formed.

For more help with selecting appropriate symbols, see page 321.

QUICK Practice

- 4 Which of the following is used to find out how many feet are in 7 yards? (3AF1.4)
- F $7 - 3$
 - G $7 + 3$
 - H 7×3
 - J $7 \div 3$

- 5 If $9 \times 15 \times 6 = 810$, then what is $15 \times 6 \times 9$? (3AF1.5)
- A 135
 - B 340
 - C 770
 - D 810

- 6 Which number makes this number sentence true? (3AF1.2)
- $$8 \times 9 < \square \times 4$$
- F 19
 - G 18
 - H 17
 - J 16

QUICK Review

READING HINT There are 3 feet in 1 yard.

Multiply the number of yards by the number of feet per yard to find the total number of feet.

For more help with expressing simple unit conversions, see page 347.

STRATEGY Use the Commutative Property of Multiplication.

If you change the order that numbers are multiplied, is there any change to the product?

For more help with using the Commutative Property of Multiplication, see page 161.

STRATEGY Eliminate choices first that do not make the number sentence true.

Replace the box with each answer choice to see which one results in a true number sentence.

For more help with solving inequalities, see page 139.

Practice on Your Own

Standard Set 1.0

- 7** The choir is set up in 3 rows with 12 singers in each row. Which number sentence shows how to find the total number of singers in the choir? **(3AF1.1)**

A $12 \times 3 = \square$
B $12 \div 3 = \square$
C $12 - 3 = \square$
D $12 + 3 = \square$

- 8** Which of the following is used to find out how many inches there are in 8 feet? **(3AF1.4)**

F $12 - 8$
G $12 + 8$
H $12 \div 8$
J 12×8

- 9** Which sign goes in the box to make the number sentence true? **(3AF1.3)**

$$49 \square 7 = 7$$

A $+$ **C** $\times$
B $-$ **D** $\div$

- 10** There are at least 15 students in Robbie's gym class. Which number sentence could be used to represent the number of students in his gym class? **(3AF1.1)**

F $\square > 15$
G $\square < 15$
H $\square = 15$
J $\square + 1 = 15$

- 11** If $4 \times (3 \times 12) = 144$, then what is $(4 \times 3) \times 12$? **(3AF1.5)**

A 12
B 96
C 144
D 168

- 12** There are fewer than 83 people in the cafeteria right now. Which number sentence could be used to represent the number of people in the cafeteria? **(3AF1.1)**

F $\square > 83$
G $\square < 83$
H $\square = 83$
J $\square - 1 = 83$

- 13** Which sign goes in the box to make the number sentence true? **(3AF1.3)**

$$88 \square 11 = 99$$

A $+$ **C** $\times$
B $-$ **D** $\div$

- 14** There are 27 third graders and 24 fourth graders on the playground. Which number sentence shows how to find how many more third graders there are? **(3AF1.1)**

F $27 \times 24 = \square$
G $27 \div 24 = \square$
H $27 - 24 = \square$
J $27 + 24 = \square$

Practice by Standard: Algebra and Functions

Standard Set 2.0: Students represent simple functional relationships.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** At the school bookstore, one notebook costs \$0.59. Two notebooks cost \$1.18. Three notebooks cost \$1.77. If the price of each notebook remains the same, how much would 4 notebooks cost? **(3AF2.1)**

A \$2.25 **C** \$2.36
B \$2.31 **D** \$2.48

- 2** The table shows the number of crayons needed for different numbers of students.

Crayons	
Students	Crayons
1	18
2	16
3	24

Each student gets the same number of crayons. How many crayons are needed for 5 students? **(3AF2.2)**

F 32 **H** 44
G 40 **J** 48

QUICK Review

STRATEGY Add the cost of 1 notebook to the given cost of 3 notebooks to find the cost of 4 notebooks.

You can solve this problem by finding the sum of \$1.77 and \$0.59.

For more help with solving problems involving functional relationships, see page 234.

STRATEGY Look for a pattern in the table. Then extend the pattern.

How many crayons are needed for each student? How many crayons would be needed for 5 students?

For more help with recognizing and extending linear patterns, see page 234.

QUICK Practice

- 3 If cashews cost \$3.79 per pound, how much would a 3-pound can of cashews cost? (3AF2.1)

A \$9.42 C \$10.96
B \$10.35 D \$11.37

- 4 The table shows the number of computers needed for different numbers of students in a computer class.

Computer Class	
Students	Computers
2	1
4	2
6	3

If there is the same number of students at each computer, how many computers will be needed for a class of 22 students? (3AF2.2)

F 44 H 11
G 22 J 6

- 5 If a taxi charges \$0.65 per mile, how much would a 12-mile taxi ride cost? (3AF2.1)

A \$7.80
B \$8.35
C \$10.60
D \$12.65

QUICK Review

STRATEGY Multiply the cost of 1 pound of cashews by 3 to find the cost of 3 pounds.

You can solve this problem by finding the product of \$3.79 and 3.

For more help with solving problems involving functional relationships, see page 234.

STRATEGY Look for a pattern in the table. Then extend the pattern.

How many students are seated at each computer? How many computers would be needed for 22 students?

For more help with recognizing linear patterns, see page 212. For more help with extending linear patterns, see page 234.

STRATEGY Multiply the cost per mile by 12 miles to find the total cost.

You can solve this problem by finding the product of \$0.65 and 12.

For more help with solving problems involving functional relationships, see page 234.

Practice on Your Own

Standard Set 2.0

- 6** At the school cafeteria, one carton of milk costs \$0.45. Two cartons of milk cost \$0.90. Three cartons of milk cost \$1.35. If the price of each carton remains the same, how much would 6 cartons of milk cost? **(3AF2.1)**

F \$2.55
G \$2.70
H \$2.85
J \$2.90

- 7** If pears cost \$0.85 per pound at a farmer's market, how much would 4 pounds of pears cost? **(3AF2.1)**

A \$3.25
B \$3.40
C \$3.80
D \$3.95

- 8** The table shows the number of ride tickets needed for different numbers of students at a carnival.

Ride Tickets	
Number of Students	Number of Tickets
1	10
2	20
3	30

If each student gets the same number of ride tickets, how many tickets are needed for 29 students? **(3AF2.2)**

F 60 **H** 250
G 180 **J** 290

- 9** If an electrician charges \$35 per hour to work on your home, how much would a 3-hour job cost? **(3AF2.1)**

A \$70
B \$85
C \$100
D \$105

- 10** Ana collects stickers in a book. The table shows the total number of stickers on different numbers of pages.

Sticker Collection	
Pages	Stickers
1	15
2	30
3	45

If each page has the same number of stickers, how many stickers are there total on 8 pages? **(3AF2.2)**

F 120
G 135
H 150
J 180

- 11** Nate's heart beats 75 times each minute. How many times will his heart beat in 7 minutes? **(3AF2.1)**

A 475
B 515
C 525
D 595

Practice by Standard: Measurement and Geometry

Standard Set 1.0: Students choose and use appropriate units and measurement tools to quantify the properties of objects.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1 Which of the following objects is lighter than 1 pound? (3MG1.1)



Desk



Dog



Skateboard



Pencil

QUICK Review

STRATEGY Eliminate the answer choices that are unreasonable.

Think of how heavy or light 1 pound is. About how much does each object weigh?

For more help with estimating the weights of objects, see page xxx.

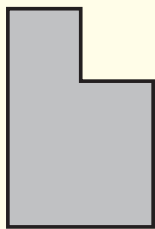


Didn't find a grayscale soccer ball with this art spec number.

QUICK Practice

- 2 What is the area of this figure?

(3MG1.2)



= 1 square unit

- F 3 square units
- G 4 square units
- H 5 square units
- J 6 square units

- 3 A playground is shaped like a rectangle 30 yards long and 20 yards wide.



20 yards

What is the perimeter, in yards, of the playground? (3MG1.3)

- A 120 yards C 60 yards
- B 100 yards D 50 yards

QUICK Review

READING HINT The *area* of a figure is the amount of space that it covers.



How many unit squares would it take to completely cover the figure?

For more help with finding the area of a figure, see page 428.

READING HINT The *perimeter* is the distance around the outside of a figure.

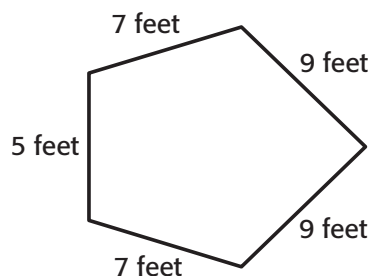
What are the lengths of the 4 sides?
What is the sum of these lengths?

For more help with finding the perimeter of a figure, see page 422.

Practice on Your Own

Standard Set 1.0

- 4** What is the perimeter of the figure?
(3MG1.3)

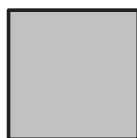


- F** 37 feet **H** 43 feet
G 39 feet **J** 47 feet

- 5** There are 60 minutes in each hour. How many minutes are there in a 7-hour school day? (3MG1.4)

- A** 360 minutes
B 380 minutes
C 420 minutes
D 440 minutes

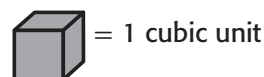
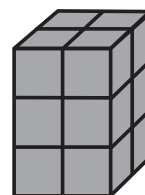
- 6** The square below has 3-centimeter sides. What is the area of the square?
(3MG1.2)



 = 1 square centimeter

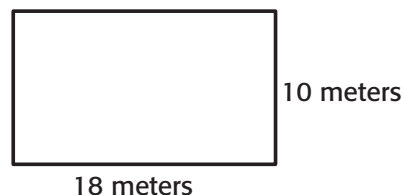
- F** 3 square centimeters
G 6 square centimeters
H 7 square centimeters
J 9 square centimeters

- 7** What is the volume of the figure below? (3MG1.2)



- A** 6 cubic units
B 8 cubic units
C 10 cubic units
D 12 cubic units

- 8** Edgar's backyard has the size and shape shown below. It is shaped like a rectangle.



Edgar wants to place fencing around the perimeter of the yard. How much fencing is needed? (3MG1.3)

- F** 28 meters
G 56 meters
H 72 meters
J 80 meters

Practice by Standard: Measurement and Geometry

Standard Set 2.0: Students describe and compare the attributes of plane and solid geometric figures and use their understanding to show relationships and solve problems.

DIRECTIONS

Choose the best answer.

QUICK Practice

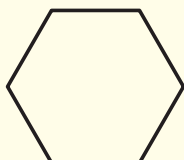
1 Which of these is a pentagon?

(3MG2.1)

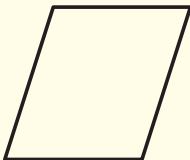
A



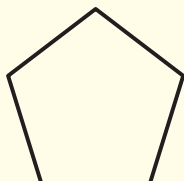
C



B



D



2 An equilateral triangle **MUST** have

(3MG2.2)

- F** no sides that are the same length.
- G** 2 sides that are the same length.
- H** 3 sides that are the same length.
- J** 4 sides that are the same length.

QUICK Review

READING HINT A pentagon is a figure with 5 sides?

Which figure has 5 sides?

For more help with identifying polygons, see page 419.

STRATEGY Eliminate answer choices that are unreasonable.

How would you describe the sides of an equilateral triangle? Which answer choice matches this description?

For more help with identifying the attributes of triangles, see page 434.

QUICK Practice

3 How many equal sides are there in a square? (3MG2.3)

- A** 4
- B** 3
- C** 2
- D** none

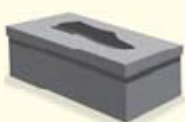
QUICK Review

STRATEGY Eliminate choices first that you know are not correct.

Draw a square on your paper. How many equal sides does it have?

For more help with identifying the attributes of quadrilaterals, see page 438.

4 Which object is a cone? (3MG2.5)

F**H****G****J**

STRATEGY Eliminate answer choices that are not shaped like a cone.

What does a cone look like? Which object is shaped like this?

For more help with identifying three-dimensional figures, see page 444.

5 How many equal sides must a triangle have to be isosceles? (3MG2.2)

- A** none
- B** 1
- C** 2
- D** 3

STRATEGY Eliminate answer choices that do not match the definition of an isosceles triangle.

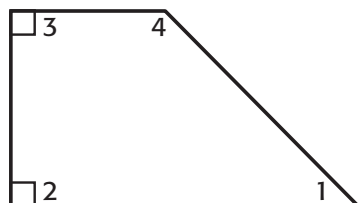
How would you describe the sides of an isosceles triangle? Which answer choice matches this description?

For more help with identifying the attributes of triangles, see page 434.

Practice on Your Own

Standard Set 2.0

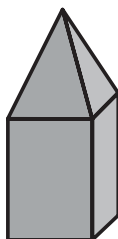
- 6** Look at the angles in the figure below.



Which angle has a measure that is greater than the measure of a right angle? (3MG2.4)

- F** angle 1 **H** angle 3
G angle 2 **J** angle 4

- 7** Which shapes make up this solid object? (3MG2.6)



- A** prism and cone
B prism and pyramid
C pyramid and cone
D prism and cube

- 8** A rectangle MUST have (3MG2.3)

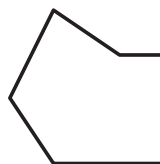
- F** five sides.
G four equal sides.
H three equal sides.
J four right angles.

- 9** One side of a parallelogram is 15 inches long. Another side of the parallelogram is 10 inches long. What are the lengths of the other two sides of the parallelogram? (3MG2.3)

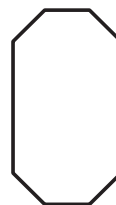
- A** 10 inches and 15 inches
B 10 inches and 10 inches
C 15 inches and 15 inches
D 11 inches and 13 inches

- 10** Which of these is an octagon? (3MG2.1)

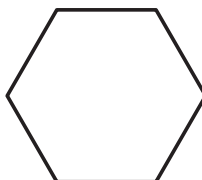
F



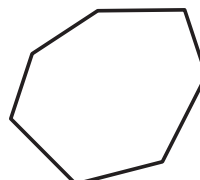
H



G



J



- 11** How many right angles does a right triangle have? (3MG2.2)

- A** 0 **C** 2
B 1 **D** 3

- 12** A hexagon has how many sides? (3MG2.1)

- F** four
G five
H six
J eight

Practice by Standard: Statistics, Data Analysis, and Probability

Standard Set 1.0: Students conduct simple probability experiments by determining the number of possible outcomes and make simple predictions.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** Missy rolled a number cube 12 times and recorded the results. The cube landed on an odd number 5 times, an even number 7 times, and a number greater than two 8 times. Which tally chart shows these results?

(3SDAP1.2)

A

Roll Results	
Odd Number	
Even Number	
Greater than 2	

B

Roll Results	
Odd Number	
Even Number	
Greater than 2	

C

Roll Results	
Odd Number	
Even Number	
Greater than 2	

D

Roll Results	
Odd Number	
Even Number	
Greater than 2	

QUICK Review

STRATEGY Eliminate answer choices that do not show the correct tallies.

Which chart has 5 tally marks for an odd number, 7 tally marks for an even number, and 8 tally marks for a number greater than two?

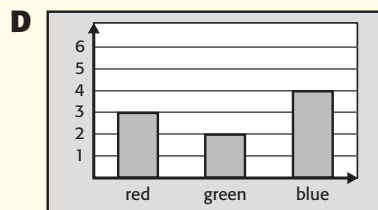
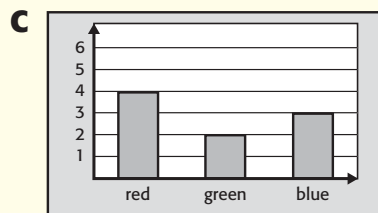
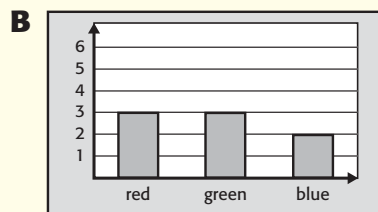
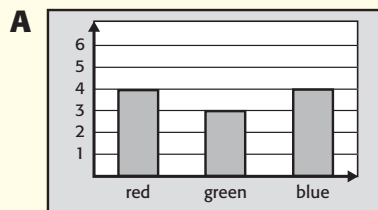
For more help with recording the outcomes of an experiment systematically, see page 473.

QUICK Practice

- 2 Andy has 1 penny, 3 nickels, 5 dimes, and 4 quarters. He closes his eyes and picks one coin without looking. Is it certain, likely, unlikely, or impossible that he will pick a penny? (3SDAP1.1)

F certain H unlikely
G likely J impossible

- 3 Rex spun a spinner 9 times. It landed on red 3 times, green twice, and blue 4 times. Which bar graph shows these results? (3SDAP1.3)



QUICK Review

READING HINT An event is *certain* if it will always happen and *impossible* if it will never happen. Otherwise it is either *likely* or *unlikely*.

How many coins are in the piggy bank?
How many of them are pennies? How many of them are not pennies?

For more help with determining the likelihood of an event, see page 484.

STRATEGY Eliminate answer choices that do not show the correct number for each spin.

Which bar graph shows the correct number of spins for each spin?

Recall that there are many ways to display data.

Color Spin	
Color	Tally
Red	
Green	
Blue	

For more help with displaying the results of an experiment, see page 493.

Practice on Your Own

Standard Set 1.0

- 4** The tally chart shows the weights of the fish that Josie caught on a week-long fishing trip.

Weight (lb)	Tally
1	
2	
3	
4	
5	

Which graph shows these results?

(3SDAP1.3)

F



G



H



J



- 5** At a carnival game, there are 20 rubber ducks floating on the water. On the bottom of 15 of the ducks, there is a black dot. On the other 5 ducks there is no dot. Players pick a duck and win a prize if it has a dot on the bottom. Is it certain, likely, unlikely, or impossible that a player will win a prize? (3SDAP1.1)

- A** certain **C** unlikely
B likely **D** impossible

- 6** To play a math game, players take turns selecting a chip from a bag. They note the color and return the chip to the bag. The tally chart shows how many times each color has been selected after 30 turns.

Color	Tally
Black	
White	
Purple	
Green	

Which color chip will the next player most likely select? (3SDAP1.4)

- F** black **H** purple
G white **J** green

- 7** Refer to the tally chart in Exercise 6. Is it certain, likely, unlikely, or impossible that the next player will select an orange chip? (3SDAP1.4)

- A** certain **C** unlikely
B likely **D** impossible

Practice by Standard: Mathematical Reasoning

Standard Set 1.0: Students make decisions about how to approach problems.

DIRECTIONS

Choose the best answer.

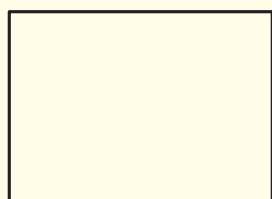
QUICK Practice

- 1** During the first four days of vacation, Carlo found 3, 5, 7, and 9 seashells. Suppose this pattern continues. How many seashells will he find on the fifth day?

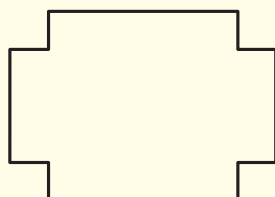
(3MR1.1)

- A** 9 **C** 11
B 10 **D** 13

- 2** Mira is creating a picture frame with a 10-inch by 12-inch board. The perimeter is 44 inches. She wants to cut out a 2-inch square 2 inches long from each corner. What will the new perimeter be? (3MR1.2)



$P = 44$ inches



$P = ?$

- F** 48 inches **H** 40 inches
G 44 inches **J** 32 inches

QUICK Review

STRATEGY Look for a pattern in the problem statement.

How many more shells does Carlo find each day?

For more help with observing and extending patterns, see page 17.

STRATEGY Break the problem into simpler parts. What happens in each corner of the frame?

How is the perimeter changed when one corner is removed? How will it change when all four corners are removed?

For more help with breaking a problem into simpler parts, see page 432.

Practice on Your Own

Standard Set 1.0

- 3** At the bookstore at Josey's school, one folder costs \$0.20. Two folders cost \$0.40. Three folders cost \$0.60. Josey wants to buy 10 folders. If the price of each folder remains the same, how much would 10 folders cost? **(3MR1.1)**

A \$2.00
B \$2.10
C \$2.20
D \$2.40

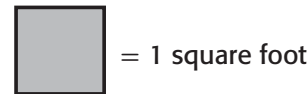
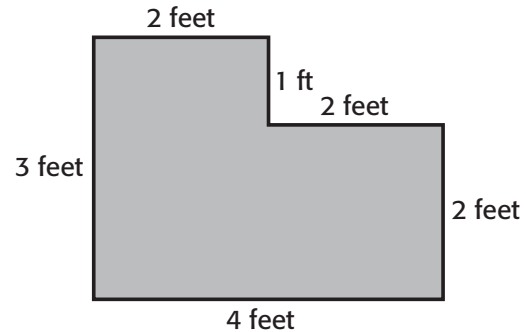
- 4** Reggie collects baseball cards. He keeps them in a binder. The table shows the total number of cards for different numbers of pages.

Baseball Card Collection	
Pages	Cards
1	12
2	24
3	36
4	48
5	60
6	?
7	84
8	96
9	108

If each page has the same number of baseball cards, how many cards total are there on 6 pages? **(3MR1.1)**

F 64
G 68
H 70
J 72

- 5** What is the area of the figure below? **(3MR1.2)**



A 8 square feet
B 10 square feet
C 11 square feet
D 12 square feet

- 6** Elio bought 6 packages of hot dog buns for a party. There are 8 buns in each package. They sell for \$1.59 per package. How much did Elio spend on hot dog buns? **(3MR1.1)**

F \$12.72 **H** \$9.54
G \$10.56 **J** \$9.28

- 7** A gift shop sells one candle for \$4, two candles for \$8, and three candles for \$12. Danielle wants to buy nine candles. If the price of each candle remains the same, how much would nine candles cost? **(3MR1.1)**

A \$2.00 **C** \$2.20
B \$2.10 **D** \$2.40

Practice by Standard: Mathematical Reasoning

Standard Set 2.0: Students use strategies, skills, and concepts in finding solutions.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** The table shows the number of fans at the Badgers' first 4 soccer games. Estimate the total number of people for these 4 games.

(3MR2.1)

Badgers Soccer	
Game	People
1	79
2	83
3	82
4	78

- A** 250 **C** 300
B 280 **D** 320

- 2** To ride a roller coaster, you must be at least 48 inches tall. Which the following shows the heights of people who can ride the roller coaster? (3MR2.3)

- F** height = 48
G height + 1 = 48
H height > 48
J height < 48

QUICK Review

STRATEGY Use rounding to estimate the solution.

Each number is close to 80. You can estimate the total number of people by finding 4×80 .

For more help with estimating solutions, see page 70.

READING HINT The phrase *at least* means greater than or equal to.

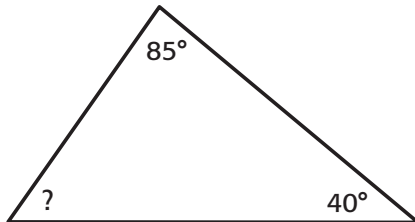
Can you ride the roller coaster if you are 48 inches tall? Can you ride the roller coaster if you are taller than 48 inches?

For more help with using numbers and symbols to represent mathematical reasoning, see pages 138 and 320.

Practice on Your Own

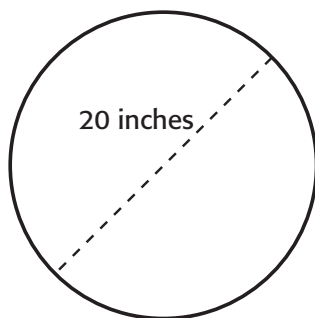
Standard Set 2.0

- 3** The sum of the angles of a triangle is 180° . Look at the triangle below. What is the measure of the missing angle? (3MR2.6)



- A** 45°
- B** 55°
- C** 65°
- D** 75°

- 4** Melisa estimates how far a wheel will roll in 1 turn by multiplying the size of the wheel by 3. About how far would a 20-inch bicycle wheel in 1 turn? (3MR2.2)



- F** 20 inches
- G** 40 inches
- H** 60 inches
- J** 80 inches

- 5** The third graders are raising money for a charity. The table shows how much has been raised during the first 4 months.

Fundraiser	
Month	Money Raised
1	\$322
2	\$409
3	\$353
4	\$328

How much money has been raised so far? Round your answer to the nearest \$10. (3MR2.5)

- A** \$1380
- B** \$1400
- C** \$1410
- D** \$1450

- 6** There are 140 students in 5 classrooms. Each room has the same number of students in it. How could you find how many students are in each classroom? (3MR2.4)

- F** Add 5 to 140.
- G** Subtract 5 from 140.
- H** Multiply 140 by 5.
- J** Divide 140 by 5.

- 7** Gerald has four bags of bagels. Each bag has 16 bagels. How could you find the total number of bagels in the four bags? (3MR2.4)

- A** Add 4 and 16.
- B** Subtract 4 from 16.
- C** Multiply 4 by 16.
- J** Divide 16 by 4.

Practice by Standard: Mathematical Reasoning

Standard Set 3.0: Students move beyond a particular problem by generalizing to other situations.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1 The table shows how many players and winners there were at a carnival game the first 3 days.

Dart Game	
Players	Winners
80	20
60	15
120	30

Suppose there will be 100 players tomorrow. Which is the most reasonable number of winners to expect? (3MR3.1)

- A** 15 **C** 25
B 20 **D** 30

- 2 Irena drew the following pattern.

1 1 1
 1 2 2 1
 1 3 3 3 1
 1 4 4 4 4 1

What is the next number in the pattern? (3MR3.2)

- F** 1 5 5 1 **H** 1 5 5 5 5 1
G 1 5 5 5 1 **J** 1 5 5 5 5 5 1

QUICK Review

STRATEGY Look for a common fraction of winners to players during the first 3 days. Then choose the most reasonable answer.

What fraction of the players were winners during each of the first 3 days? What would be an equivalent fraction of 100 players?

For more help with evaluating the reasonableness of a solution, see page 120.

STRATEGY Look for a pattern and predict the next number.

What happens to the numbers between the 1s with each term?

For more help with extending patterns, see page 212.

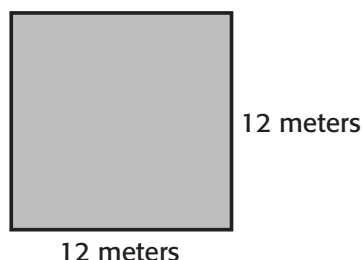
Practice on Your Own

Standard Set 3.0

- 3** Leo uses the method shown below to find the areas of squares with different side lengths.

Side Length (units)	Method	Area (square units)
1	1×1	1
2	2×2	4
3	3×3	9
4	4×4	16
5	5×5	25
6	6×6	36

What is the area of the square shown below? (3MR3.2)



- A** 156 square meters
B 144 square meters
C 132 square meters
D 96 square meters
- 4** Refer to Exercise 3. How could you use Leo's method to find the area of a square with a side length of 15 inches? (3MR3.2)
- F** Find $15 + 15$.
G Find $15 - 140$.
H Find 15×15 .
J Find $15 \div 15$.

- 5** The table shows how many days it has rained over the past 4 months.

Month	Rainy Days
April	13
May	12
June	10
July	9

Which is the most reasonable number of rainy days to expect for the month of August? (3MR3.1)

- A** 15
B 13
C 8
D 5

- 6** Tasha has kept track of the amount of savings over the last few months.

Tasha's Savings	
Months	Total Savings
1	\$35
2	\$70
3	\$105
4	\$140

Suppose Tasha continues to save the same amount each month. How much will she have saved after 12 months? (3MR3.3)

- F** \$420
G \$390
H \$385
J \$375